

IT-SIMPLERA INSTITUTE

INTERNSHIP WEEKLY REPORT

Data Analysis & Business Understanding — Week 1

Online Retail II — Exploratory Data Analysis Report

ITSimplera Institute | AI & ML Internship | Week 1

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Programme	AI & ML Internship — 8-Week Remote (through August 2026)
Task	Week 1 — Data Analysis & Business Understanding
Submission Date	Saturday, 4 July 2026
Dataset	Online Retail II — UCI ML Repository (2009–2011)
Tools Used	Python 3, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

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Executive Summary

This report presents a complete Exploratory Data Analysis (EDA) of the Online Retail II dataset (UCI Machine Learning Repository), covering real transaction data from a UK-based online retailer spanning December 2009 to December 2011. The analysis was produced as the Week 1 Task of the ITSimplera AI & ML Internship programme.

The objective was to help a retail company understand customer purchase behaviour, identify top-selling products, and determine which countries generate the most revenue — ahead of investing in a recommendation system. This is an observation-only EDA: no rows were removed or modified at any stage.

Total Rows	Unique Customers	Unique Invoices	Unique Products	Countries	Total Revenue
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Submission Details

1,067,371	5,942	53,628	5,305	43	GBP 19,287,251
Student Name	Farhan Khan				
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Programme	AI & ML Internship — 8-Week Remote (through August 2026)				
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Dataset at a Glance

Section 1 Dataset Overview

The Online Retail II dataset contains real transactional records from a UK-based non-store online retailer. It is distributed as an Excel workbook with two sheets — one per fiscal year — which were concatenated into a single working dataframe of 1,067,371 rows for this analysis.

1.1 Dataset Structure

Attribute	Detail
Source	UCI ML Repository — Online Retail II
File Format	Excel Workbook (.xlsx) — 2 sheets
Sheet 1 (Year 2009-2010)	525,461 rows
Sheet 2 (Year 2010-2011)	541,910 rows
Combined Shape	1,067,371 rows x 8 columns
Date Range	1 December 2009 — 9 December 2011
Primary Market	United Kingdom (84.94% of revenue)
International Markets	42 additional countries

1.2 Column Descriptions

Column	Type	Description
Invoice	object	Invoice number. Prefix "C" = cancellation/return.
StockCode	object	Unique product identifier code.
Description	object	Product name. Has 4,382 (0.41%) missing values.

Quantity	int64	Units per line item. Negative values = returns.
InvoiceDate	datetime64[us]	Date and time of the transaction.
Price	float64	Unit price in GBP (Great British Pounds).
Customer ID	float64	Customer identifier. 22.77% missing (anonymous users).
Country	str	Country of the customer placing the order.

Section 2 — Data Quality Assessment

Before any analysis, all data quality issues were identified and documented. No data was removed or modified — this entire EDA is observation-only, in line with the task instructions. Cleaning decisions are deferred to the next phase.

2.1 Missing Values

Column	Missing Count	Missing %	Business Implication
Description	4,382	0.41%	Minor — likely data-entry omissions for special adjustment lines.
Customer ID	243,007	22.77%	Critical — anonymous/guest transactions. Cannot be used for customer-level recommendation modelling.

2.2 Duplicate Rows

Metric	Count	Percentage of Dataset
Fully duplicated rows (all 8 columns identical)	34,335	3.22%

Duplicate rows likely arise from system re-imports or data pipeline issues. They should be deduplicated before any downstream modelling, but are kept as-is here per task requirements.

2.3 Additional Anomalies (Observation Only)

Anomaly	Count	Percentage	Notes
Negative Quantity (returns)	22,950	2.15%	Legitimate returns. Invoice usually starts with "C".
Cancelled Invoices (prefix "C")	19,494	1.83%	Full cancellations — should be excluded from sales analysis.
Rows with Price = 0	6,202	0.58%	Free samples, manual adjustments, or data-entry errors.
Missing Customer ID	243,007	22.77%	Anonymous purchasers — cannot be linked to a customer profile.

Key takeaway: The 22.77% missing Customer ID rate is the most impactful data quality issue for the planned recommendation system. Over 243,000 transactions cannot be attributed to an identifiable customer. These records would need to be excluded or handled via invoice-level session grouping before any collaborative filtering or customer-level model can be built.

Section 3 Top 10 Best-Selling Products

Products were ranked by two separate metrics: total quantity sold (units) and total revenue generated (GBP). The two rankings differ meaningfully — high-volume items (inexpensive novelty and party goods) are not the same as high-revenue items (premium homewares). This has direct implications for how a recommendation system should score and rank products.

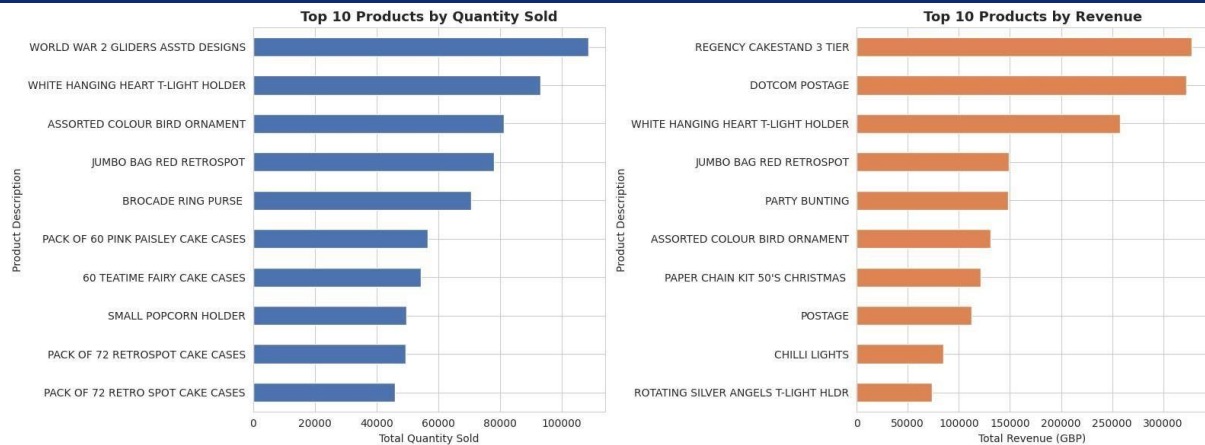


Figure 1 — Top 10 Products by Quantity Sold (left) and by Revenue Generated in GBP (right)

3.1 Top 10 Products by Quantity Sold

Rank	Product Description	Total Quantity Sold
1	WORLD WAR 2 GLIDERS ASSTD DESIGNS	108,545
2	WHITE HANGING HEART T-LIGHT HOLDER	93,050
3	ASSORTED COLOUR BIRD ORNAMENT	81,306
4	JUMBO BAG RED RETROSPOT	78,090
5	BROCADE RING PURSE	70,700
6	PACK OF 60 PINK PAISLEY CAKE CASES	56,575
7	60 TEATIME FAIRY CAKE CASES	54,366
8	SMALL POPCORN HOLDER	49,616
9	PACK OF 72 RETROSPOT CAKE CASES	49,344
10	PACK OF 72 RETRO SPOT CAKE CASES	46,106

3.2 Top 10 Products by Revenue (GBP)

Rank	Product Description	Total Revenue (GBP)
1	REGENCY CAKESTAND 3 TIER	GBP 327,813.65
2	DOTCOM POSTAGE	GBP 322,647.47
3	WHITE HANGING HEART T-LIGHT HOLDER	GBP 257,533.90
4	JUMBO BAG RED RETROSPOT	GBP 148,800.64
5	PARTY BUNTING	GBP 147,948.50
6	ASSORTED COLOUR BIRD ORNAMENT	GBP 131,413.85

7	PAPER CHAIN KIT 50'S CHRISTMAS	GBP 121,662.14
8	POSTAGE	GBP 112,341.00
9	CHILLI LIGHTS	GBP 84,854.16
10	ROTATING SILVER ANGELS T-LIGHT HLDR	GBP 73,814.72

Insight: Only 3 products appear in both top-10 lists (WHITE HANGING HEART T-LIGHT HOLDER, JUMBO BAG RED RETROSPOT, ASSORTED COLOUR BIRD ORNAMENT). This shows that popularity by volume and value contribution are different dimensions. A robust recommendation system should use a hybrid ranking score combining both purchase frequency and revenue contribution, rather than optimising for popularity alone.

Section 4 Sales Performance by Country

The dataset spans 43 countries. The United Kingdom is overwhelmingly dominant, accounting for 84.94% of total revenue (GBP 16,382,584). The second chart (Figure 3) excludes the UK to make the international market distribution visible.

4.1 Revenue by Country Chart

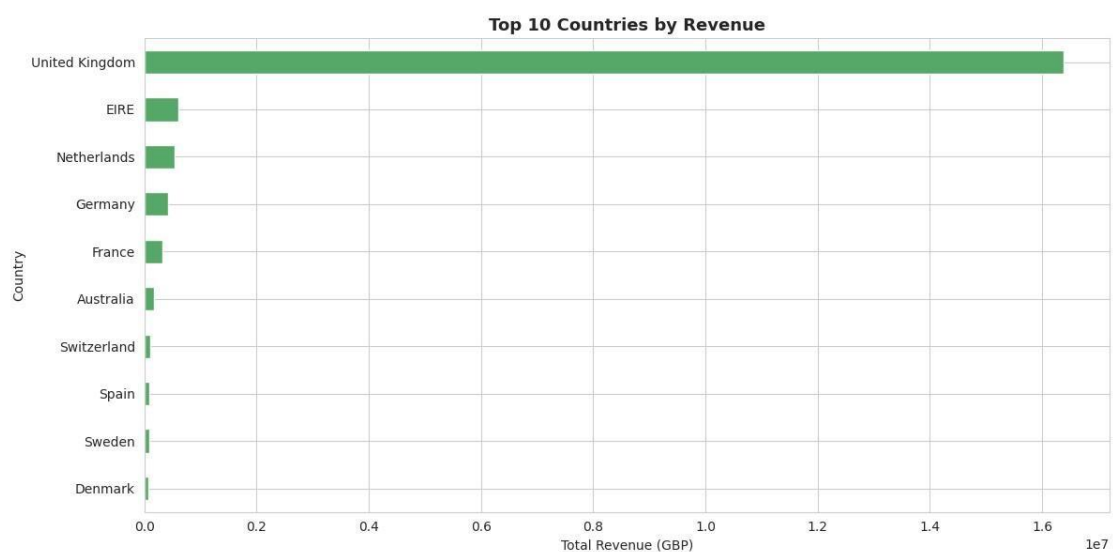


Figure 2 — Top 10 Countries by Total Revenue (GBP) — including United Kingdom

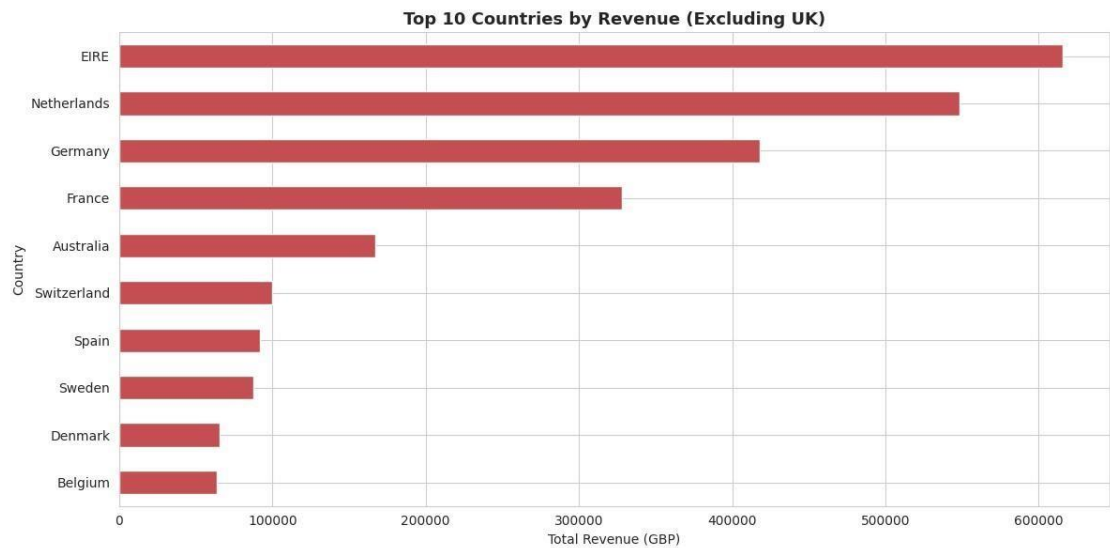


Figure 3 — Top 10 Countries by Revenue (excluding UK) — international market breakdown

4.2 Country Summary Table — Top 10

Country	Revenue (GBP)	Qty Sold	Invoices	Customers	Rev Share
United Kingdom	GBP 16,382,584	8,692,875	49,108	5,410	84.94%
EIRE	GBP 615,520	331,341	806	5	3.19%
Netherlands	GBP 548,525	381,951	250	23	2.84%
Germany	GBP 417,989	224,581	1,095	107	2.17%
France	GBP 328,192	184,952	746	95	1.70%
Australia	GBP 167,129	103,706	117	15	0.87%

Switzerland	GBP 99,729	52,378	123	22	0.52%
Spain	GBP 91,859	45,156	188	41	0.48%
Sweden	GBP 87,809	87,875	129	19	0.46%
Denmark	GBP 65,741	235,218	53	12	0.34%

Notable observation: EIRE (Ireland) has only 5 unique customers but 806 invoices and GBP 615,520 in revenue — suggesting a small number of large-volume B2B buyers. Denmark similarly has high quantity (235,218 units) relative to customers (12) and revenue, indicating wholesale-pattern purchasing. These outlier customers would distort a recommendation system if not segmented separately from standard retail customers.

Section 5 Revenue Over Time (Monthly Trend)

Revenue was aggregated by calendar month across both years. A clear seasonal pattern emerges with peaks in October–November each year (driven by pre-holiday shopping) followed by a sharp January dip. The overall revenue level is higher in 2011 vs 2010, suggesting year-on-year growth.

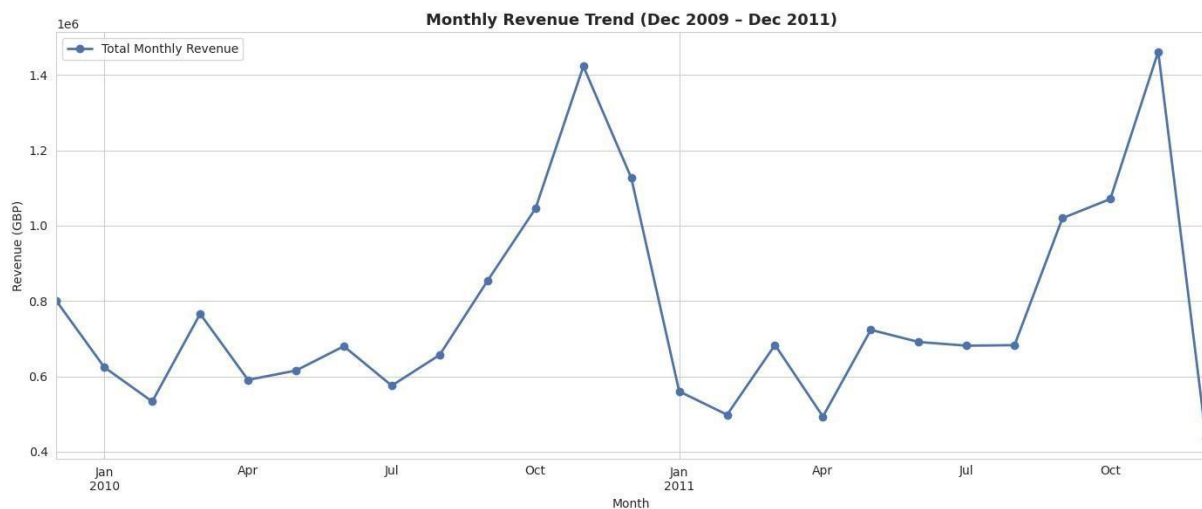


Figure 4 — Monthly Revenue Trend, December 2009 to December 2011

Key Observations

- **Consistent Q4 peaks:** Both 2010 and 2011 show strong revenue spikes in October–November, confirming reliable seasonal demand ahead of the holiday period.
- **January slowdowns:** Revenue drops sharply in January both years — a classic post-holiday retail pattern.

- **Year-on-year growth:** 2011 peaks appear higher than corresponding 2010 months, suggesting the business was growing during this period.
- **Incomplete December 2011:** Data ends 9 December 2011, so the final month appears lower than it would be with a full month of transactions.
- **Implication for recommender:** The recommendation system should incorporate seasonality (e.g., timebased features) to avoid recommending out-of-season products.

Section 6 Correlation Heatmap

A Pearson correlation matrix was computed for the four numeric columns: Quantity, Price, Revenue, and Customer ID. This identifies linear associations between features and highlights which variables move together — relevant for feature selection in the next modelling phase.



Figure 5 — Pearson Correlation Heatmap of Numeric Features

Correlation Matrix

	Quantity	Price	Revenue	Customer ID
Quantity	1.000	-0.001	0.763	-0.005
Price	-0.001	1.000	0.061	-0.004
Revenue	0.763	0.061	1.000	-0.003
Customer ID	-0.005	-0.004	-0.003	1.000

Interpretation

- **Quantity vs Revenue ($r = 0.763$):** Strong positive correlation — as expected since $\text{Revenue} = \text{Quantity} \times \text{Price}$. Order volume is the primary driver of total revenue.
- **Price vs Revenue ($r = 0.061$):** Weak correlation — unit price alone is a poor predictor of transaction revenue. Most revenue variation comes from quantity ordered, not price per unit.

- **Customer ID (all correlations ≈ 0):** No meaningful correlation with numeric fields, as expected for an identifier column assigned sequentially.

Implication: Quantity should be weighted more heavily than Price as a feature in any revenue-based recommendation scoring model.

Section 7 Outlier Detection

Box plots were produced for Quantity, Price, and Revenue. Outliers were quantified using the standard IQR rule: any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is flagged as an outlier. All three fields exhibit heavy tails on both positive and negative sides.

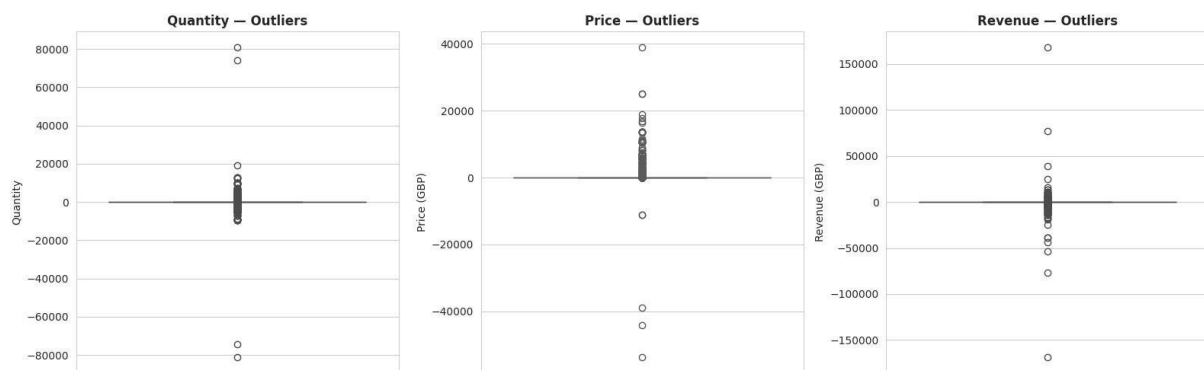


Figure 6 — Box Plots showing outlier extent for Quantity, Price, and Revenue

Outlier Summary (IQR Method)

Field	Outlier Count	% of Rows	Nature of Extremes
Quantity	116,489	10.91%	Very large bulk orders (wholesale buyers); large negative values (returns/cancellations)
Price	68,105	6.38%	Very high unit prices for premium goods; small number of zero-price entries
Revenue	90,922	8.52%	Driven by combination of high-quantity bulk orders and high-price items

Important: These outliers are legitimate business events, not data errors. Wholesale buyers genuinely place orders of tens of thousands of units; returns genuinely produce large negative quantities. Blindly removing outliers would discard meaningful business signals. Instead, the recommendation system design should explicitly segment retail customers from wholesale/B2B customers before modelling, as their behaviour patterns are fundamentally different.

Section 8 Business Insights

The following five insights are derived directly from the EDA findings and are intended to inform the company's investment decision around a recommendation system. Each insight highlights a pattern in the data and its practical implication.

1 UK-Centric Revenue Base Requires Domestic-First Strategy

The United Kingdom accounts for 84.94% of total revenue (GBP 16.38M out of GBP 19.29M total). International markets — led by EIRE, Netherlands, Germany, and France — contribute the remaining 15.06% across 42 countries combined. The recommendation system should be built and optimised for UK purchasing behaviour first. International localisation (currency, seasonality, product preferences) should be handled in a second phase, starting with the Netherlands and Germany which have meaningful customer bases (23 and 107 customers respectively).

2 Strong Q4 Seasonality — Recommendation Engine Needs Time-Awareness

Monthly revenue consistently peaks in October–November and drops sharply in January. A recommendation system trained on raw transaction history without temporal features would fail to account for this — it might recommend Christmas-season products year-round. Time-based features (month, week, days-to-holiday) should be incorporated as contextual signals in the recommendation model.

3 Volume Leaders vs Revenue Leaders Are Different Products

Only 3 products appear in both the top-10 by quantity and top-10 by revenue. High-volume sellers are predominantly small, inexpensive novelty and party items, while high-revenue sellers are premium homewares (e.g., Regency Cakestand 3 Tier at GBP 327,813). A recommendation system optimising purely on purchase frequency would systematically under-recommend high-margin products. A hybrid relevance score — weighting both co-purchase frequency and revenue contribution — is recommended.

4 22.77% Missing Customer IDs Limits Recommendation Coverage

243,007 of 1,067,371 transactions (22.77%) have no Customer ID. These represent anonymous or guest purchases that cannot be attributed to a customer profile. Any user-based collaborative filtering approach will only cover the 77.23% of transactions with identifiable customers. To maximise coverage, the team should also implement an item-based or session-based recommendation fallback for anonymous users, using invoice-level grouping as a proxy for session.

5 Wholesale Buyers Distort Retail Patterns — Segmentation Required

Several countries (notably EIRE with 5 customers and 806 invoices) and many high-quantity transaction lines indicate the presence of wholesale/B2B buyers purchasing at radically different scales to retail customers. If these buyers are included in training a retail recommendation model, their extreme order patterns will dominate the signal and produce irrelevant recommendations for average retail shoppers. Customer segmentation by order size and frequency should precede any recommendation model training.

Conclusion

This Exploratory Data Analysis of the Online Retail II dataset provides a comprehensive foundation for the company's upcoming recommendation system investment. The analysis covered all required areas: dataset structure, data quality, top-product analysis, country-level performance, monthly revenue trends, correlation analysis, outlier detection, and five business insights — all without modifying the underlying data.

Summary of Key Findings

- The dataset contains 1,067,371 transactions across 43 countries, spanning December 2009 to December 2011.
- Data quality issues identified: 22.77% missing Customer IDs, 3.22% duplicate rows, 2.15% negative quantities (returns), 0.58% zero-price rows — none removed at this stage.
- Revenue is UK-dominated (84.94%), with seasonal Q4 peaks each year.
- Product rankings differ between volume and revenue — hybrid scoring needed for recommender.
- Wholesale buyers create heavy-tailed outliers requiring customer segmentation before modelling.
- The missing Customer ID issue is the single most impactful constraint on the planned recommendation system.